文献汇报

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Video Object Detection with an Aligned Spatial-Temporal Memory

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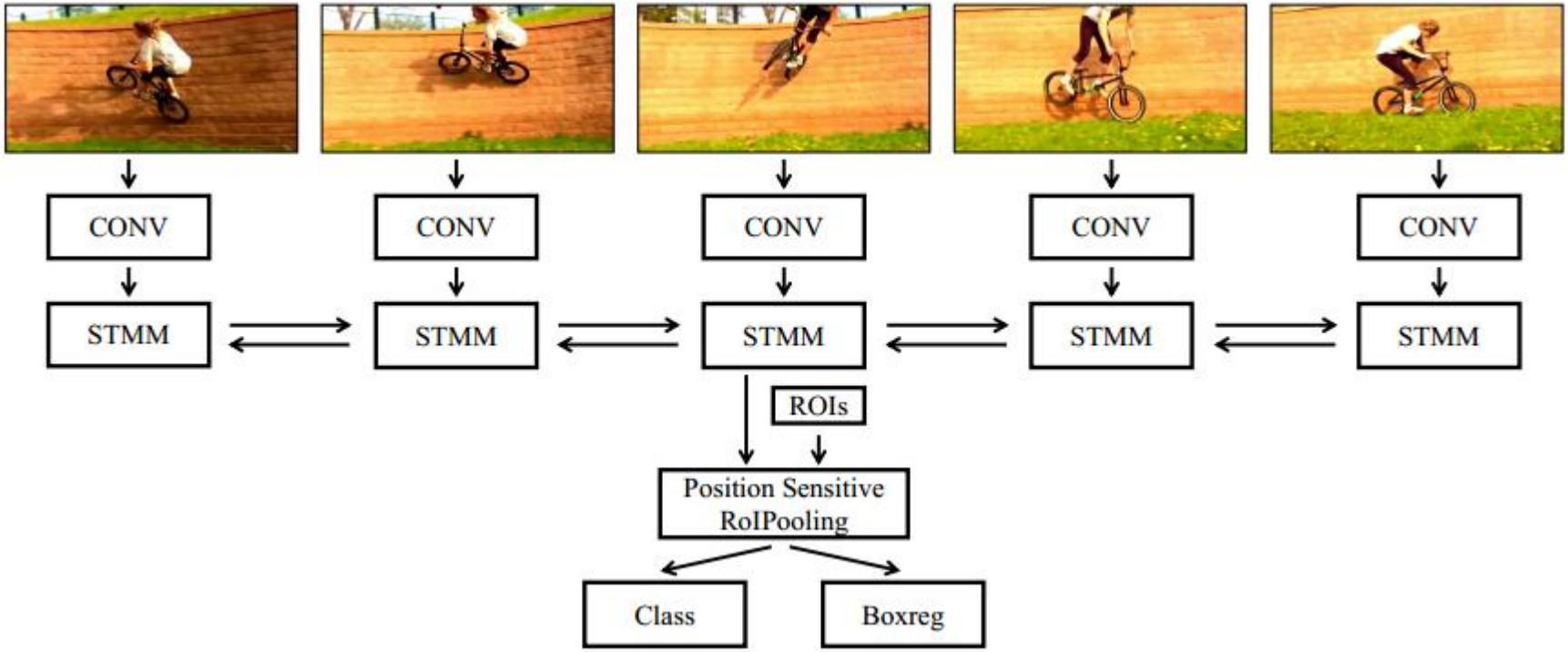


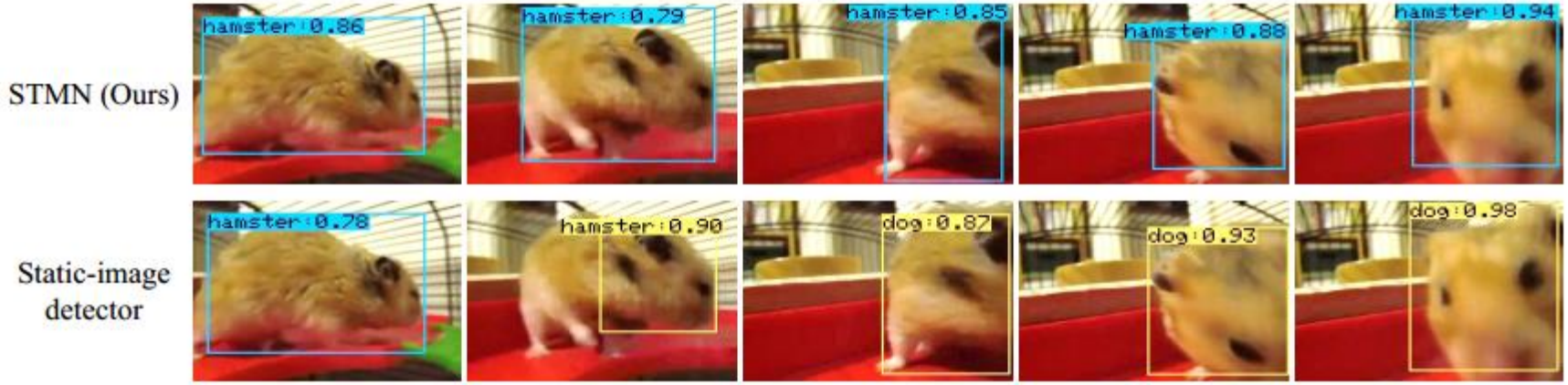
Method

视频目标检测的问题在于视频中的运动模糊、遮挡和极端视角等情况会使得静态图像检测器出问题，但是视频提供了丰富的时空和运动信息，应当学习去利用这些信息，使得视频目标检测器更加鲁棒。

FGFA等方法通过预测帧间的平移，将帧与帧之间的特征集成起来。然而这些方法处理的是固定长度窗口的图像，不能建模较长的依赖性

为了解决这些问题，我们提出Spatial-Temporal Memory Network(STMN)，用一个网络统一建模长时间的外观和运动。他的核心为Spatial-Temporal Memory Module，一种卷积递归计算，和ConvGRU基本一样。

为了实现帧间的像素级对齐，本文提出MatchTran模块显式建模帧间的平移。



STMM

$$z_t = \mathtt{BN}^*(\mathtt{ReLU}(W_z * F_t + U_z * M_{t-1})), \tag{1}$$

$$r_t = \mathtt{BN}^*(\mathtt{ReLU}(W_r * F_t + U_r * M_{t-1})), \tag{2}$$

$$\tilde{M}_t = \mathtt{ReLU}(W * F_t + U * (M_{t-1} \odot r_t)), \tag{3}$$

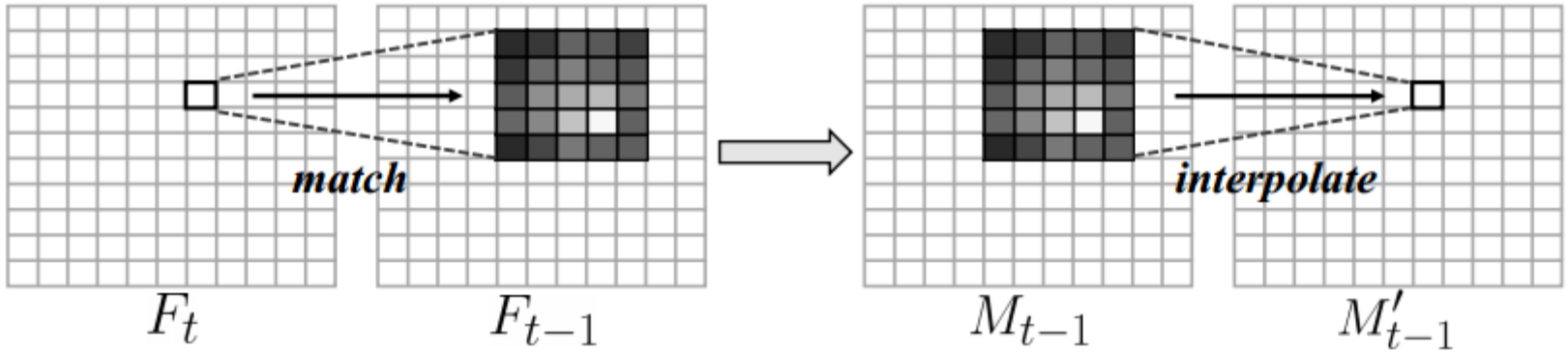
$$M_t = (1 - z_t) \odot M_{t-1} + z_t \odot \tilde{M}_t, \tag{4}$$

与ConvGRU的不同。

视频目标检测训练的过程中的一个现实的问题就是，由于视频具有冗余性，使得数据的多样化和图像数据集没法比，所以希望能够迁移图像数据集中的训练结果。

本文用R-FCN初始化网络，将Position-Sensitive ROI pooling的前一个卷积层替换为本文的STMM，由于R-FCN的激活函数ReLU的输出均为非负，而传统ConvGRU的tanh函数的输出为[-1,1]，这样是不匹配的，所以这里将GRU的激活函数改为ReLU，同时STMM的卷积层用替换出去的卷积层的参数进行初始化。

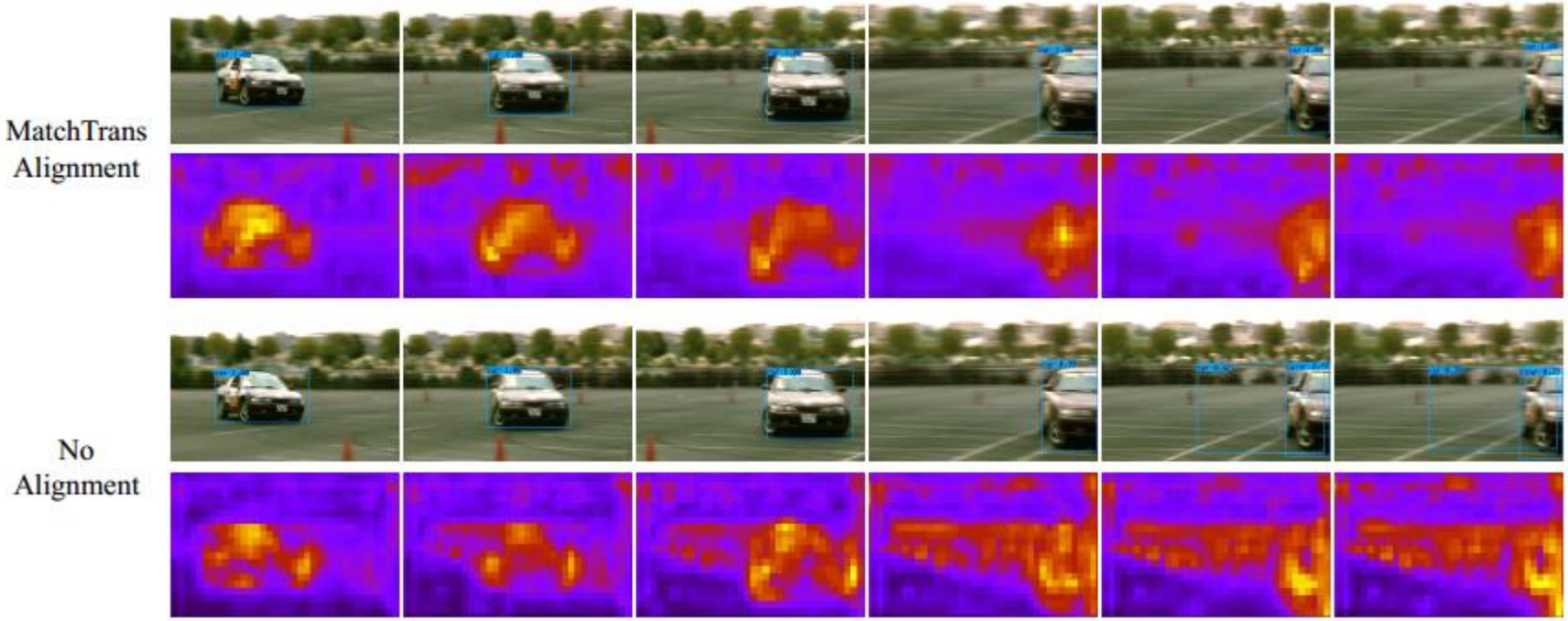
Spatial-temporal memory alignment



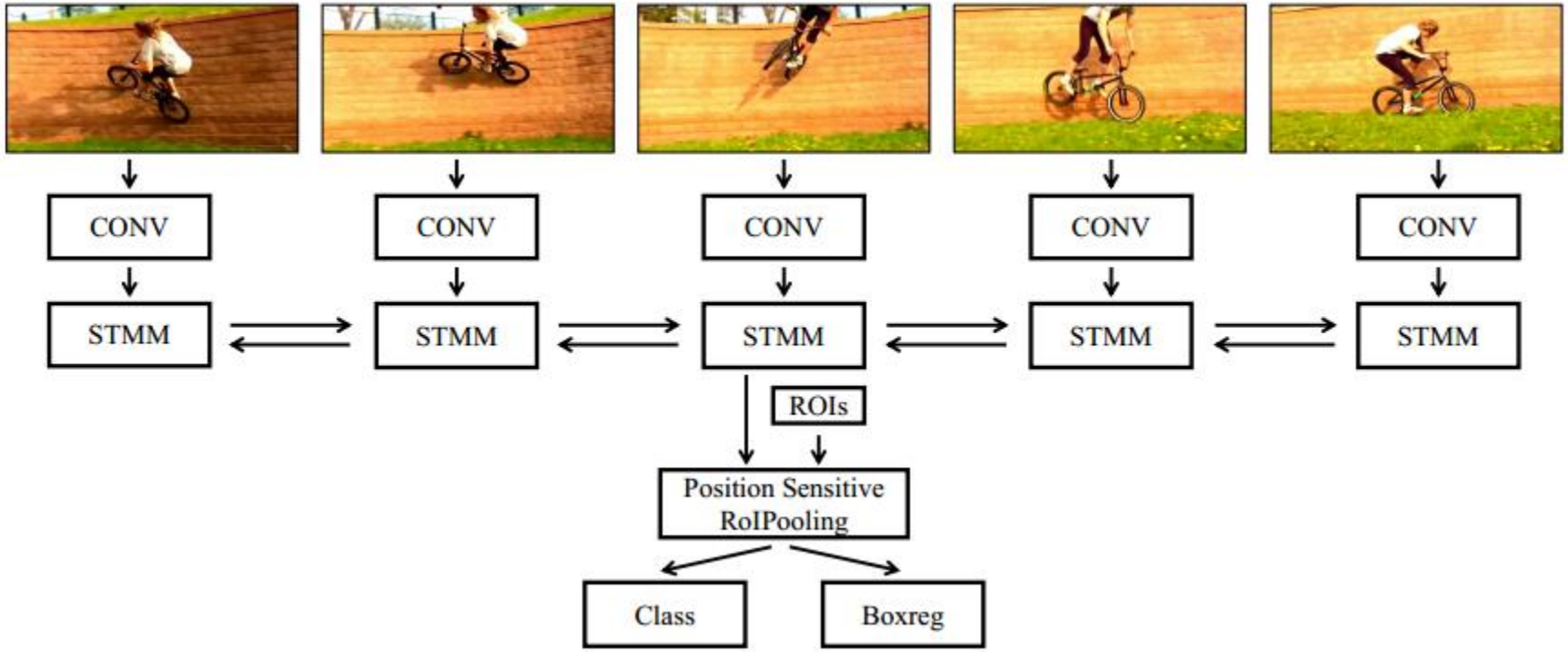
$$\Gamma_{x,y}(i,j) = \frac{F_t(x,y) \cdot F_{t-1}(x+i, y+j)}{\sum_{i,j \in \{-k,\ldots,k\}} F_t(x,y) \cdot F_{t-1}(x+i, y+j)},$$

$$M'_{t-1}(x,y) = \sum_{i,j \in \{-k,\ldots,k\}} \Gamma_{x,y}(i,j) \cdot M_{t-1}(x+i, y+j).$$

Spatial-temporal memory alignment



总体框架


CONV
CONV
CONV
CONV
CONV
STMM
STMM
STMM
STMM
STMM
ROIs
Position Sensitive
RoIPooling
Class
Boxreg

Experiments

	Base network	Base detector	Test	Val
STMN (Ours)	ResNet-101	R-FCN	-	**80.5**
D&T [14]	ResNet-101	R-FCN	-	79.8
Zhu et al. [50]	ResNet-101+DCN	R-FCN	-	78.6
FGFA [51]	ResNet-101	R-FCN	-	78.4
T-CNN [25]	DeepID+Craft [34,48]	RCNN	67.8	73.8
R-FCN [9]	ResNet-101	R-FCN	-	73.4
TPN [24]	GoogLeNet	TPN	-	68.4
STMN (Ours)	VGG-16	Fast-RCNN	**56.5**	**61.7**
Faster-RCNN [1,19]	VGG-16	Faster-RCNN	48.2	52.2
ITLab VID - Inha [1]	VGG-16	Fast-RCNN	51.5	-

Table 1. mAP comparison to the state-of-the-art on ImageNet VID. For both the "R-FCN+ResNet-101" and the "Fast-RCNN+VGG-16" settings, our STMN detector outperforms all existing methods with the same base detector and backbone network. Furthermore, in both cases, our STMN outperforms the corresponding static-image detector by a large margin.

	STMN	STMN No-MatchTrans	ConvGRU Pretrain	ConvGRU FreshFC
Test mAP	50.7	49.0	48.0	44.8

Table 2. Ablation studies on ImageNet VID. Our improvements over the baselines show the importance of memory alignment across frames with MatchTrans (vs. STMN-No-MatchTrans), and the effectiveness of using pre-trained weights with STMM over standard ConvGRU (vs. ConvGRU-Pretrain and ConvGRU-FreshFC).